

ALBANY, OH, USA

WMO: 720397

Lat: 39.211N

Lon: 82.231W

Elev: 765

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 00131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.5	(c) 9.6	(d) -5.9	(e) 4.1	(f) 8.1	(g) 0.5	(h) 5.8	(i) 13.3	(j) 23.8	(k) 33.9	(l) 20.2	(m) 37.5	(n) 2.8	(o) 270	(p) 0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	90.8	71.6	88.8	71.1	86.4	70.6	76.3	84.3	75.0	83.0	73.5	81.1	5.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.2	131.6	78.5	72.8	125.0	77.6	71.5	119.8	77.1	40.2	84.3	38.9	83.1	37.6	81.3	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	15.9	12.8	DB	-4.0	93.2	8.9	3.3	-10.4	95.6	-15.6	97.5	-20.5	99.4	-27.0	101.8
			WB	-4.7	78.1	8.3	1.9	-10.7	79.5	-15.6	80.6	-20.3	81.7	-26.4	83.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	29.5	33.2	42.9	54.2	64.1	71.3	74.1	73.0	67.2	55.1	42.9	35.9		
	DBStd	17.99	12.11	12.00	10.74	8.99	8.11	5.61	4.98	5.08	7.00	9.05	8.99	9.57		
	HDD50	2210	644	483	274	59	4	0	0	0	0	48	248	450		
	HDD65	5107	1102	891	687	337	120	13	2	3	56	329	664	903		
	CDD50	3571	7	12	54	185	441	640	748	714	517	206	35	12		
	CDD65	992	0	0	2	13	92	203	285	252	123	22	0	0		
	CDH74	9081	0	3	18	238	877	1843	2448	2205	1184	259	6	0		
	CDH80	3051	0	0	2	47	240	627	887	768	416	64	0	0		
(14) Wind	WSAvg	4.3	5.6	5.7	5.6	5.6	4.1	3.6	2.8	2.6	2.8	3.8	4.3	4.9		
(15) Precipitation	PrecAvg	41.9	3.3	2.8	4.0	3.5	4.6	4.1	4.2	3.4	3.2	2.7	2.9	3.3		
	PrecMax	50.9	6.0	5.9	7.9	5.7	8.2	9.1	5.8	5.3	10.1	7.6	5.6	8.0		
	PrecMin	33.4	1.5	0.9	1.3	1.3	0.5	0.9	2.4	1.3	0.7	0.6	1.3	1.7		
	PrecStd	5.1	1.3	1.3	1.7	1.3	2.4	1.8	0.9	1.2	2.2	1.6	1.1	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	69.4	78.5	85.6	89.6	91.4	94.9	94.7	91.4	88.1	73.0	66.0		
		MCWB	56.0	57.4	61.3	61.4	68.0	71.1	71.6	73.5	70.3	70.5	59.7	58.7		
	2%	DB	59.1	63.1	71.7	80.8	85.6	89.4	90.5	90.3	88.4	80.6	67.8	60.6		
		MCWB	53.9	54.0	57.4	60.8	68.3	70.5	72.8	71.8	70.7	64.5	56.0	55.7		
	5%	DB	53.4	57.7	66.2	76.2	82.5	86.4	88.1	87.6	85.2	76.4	63.4	55.7		
		MCWB	49.5	51.8	53.9	58.8	67.1	70.0	72.3	71.6	70.1	63.3	53.8	51.7		
	10%	DB	47.6	53.3	61.1	72.0	79.4	83.8	85.5	84.2	81.3	71.5	59.1	52.0		
		MCWB	43.3	47.0	51.3	57.4	66.1	69.1	71.7	70.5	66.6	61.4	51.6	49.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	61.0	63.0	66.3	73.7	75.8	78.5	77.8	75.3	72.3	63.2	60.7		
		MCDB	61.7	68.3	74.2	77.4	82.9	86.0	85.5	85.3	84.1	79.7	70.8	64.0		
	2%	WB	54.6	56.8	60.0	63.8	71.3	73.6	76.9	76.0	73.5	69.6	59.2	56.6		
		MCDB	58.8	60.3	67.9	72.8	79.5	82.4	84.3	83.9	82.2	76.6	63.4	59.4		
	5%	WB	49.9	53.4	57.4	61.7	69.5	72.7	75.3	74.6	72.2	66.6	55.5	53.1		
		MCDB	53.1	57.7	63.0	72.6	77.5	81.4	83.2	82.0	80.3	72.5	60.7	55.5		
	10%	WB	43.4	46.7	53.9	59.4	68.0	71.4	73.8	73.1	70.5	63.0	52.4	48.8		
		MCDB	47.3	52.0	59.1	67.9	76.3	80.1	81.1	80.0	77.2	69.4	58.6	51.8		
(35) Mean Daily Temperature Range	MDBR		17.2	18.5	21.0	24.7	21.7	21.0	19.8	21.1	22.9	22.6	21.2	15.8		
	5% DB	MCDBR	23.2	25.8	29.3	31.6	25.5	23.7	22.2	24.0	26.5	28.3	27.9	22.4		
		MCWBR	18.1	19.4	16.9	14.9	11.2	9.5	8.9	9.7	10.6	14.1	17.6	18.7		
	5% WB	MCDBR	21.3	24.7	21.8	24.7	20.1	19.8	19.1	19.2	21.2	22.1	21.3	20.3		
		MCWBR	17.9	20.1	14.4	14.0	9.5	9.3	8.8	8.5	9.3	12.3	16.0	18.1		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.324	0.352	0.384	0.427	0.448	0.465	0.464	0.395	0.365	0.328	0.302		
	taud		2.470	2.409	2.393	2.322	2.218	2.199	2.169	2.153	2.326	2.412	2.489	2.519		
	Ebn at Noon		274	285	287	283	272	265	259	256	268	266	265	269		
	Edh at Noon		25	30	35	39	45	46	47	46	36	30	25	23		
(44) All-Sky Solar Radiation	RadAvg		546	780	1127	1512	1732	1906	1816	1710	1413	994	691	470		
	RadStd		54	112	97	134	147	121	100	102	135	103	90	55		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+216	+115

Nomenclature: See separate page